

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1346

Big Sandy Peaker Plant 138 kV
December 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1346. The Interconnection Customer proposes a 389.0 MW natural gas combined cycle facility interconnecting at the Big Sandy Peaker Plant 138 kV substation in the Kentucky area. The request was received on November 30, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 332 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$33,401,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CEREDO GENERATING STA SUBSTATION – BIG SANDY PEAKER PLANT 138 KV	HUBBARDSTOWN – TRISTATE 345 KV Line	207.7	225	77.1
BIG SANDY PEAKER PLANT – N FINDLAY 138 KV	BIG SANDY PEAKER PLANT – CEREDO GENERATING STATION 138 KV	158.9	236	100.0
TRISTATE – CEREDO GENERATING STA 138 KV line	HUBBARDSTOWN SUBSTATION – TRISTATE 345 KV	157.2	223	55.7
NORTH FINDLAY – FLEWELLEN 138 KV	CEREDO GENERATING STA – BIG SANDY PEAKER PLANT 138 KV	141.6	253	100.0
FLEWELLEN – BAKER 138 KV Line	CEREDO GENERATING STA SUBSTATION – BIG SANDY PEAKER PLANT 138 KV Line	134.6	260	100.0
CEREDO GENERATING STATION – KRS II KOLOA SOLAR 138 KV	CEREDO GENERATING STA – TRISTATE 138 KV	124.7	238	69.3
VALEMOUNT 138/1 kV Autotransformer	PENCE – VALEMOUNT 345 KV Line	110.2	448	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BIG SANDY PEAKER PLANT – CEREDO GENERATING STA 138 KV	Rebuild line with higher-capacity conductor	\$3,170,000	78% (\$2,475,000)
N FINDLAY – BIG SANDY PEAKER PLANT 138 KV	Rebuild line with higher-capacity conductor	\$4,170,000	100% (\$4,170,000)
TRISTATE – CEREDO GENERATING STA 138 KV line	Rebuild line with higher-capacity conductor	\$1,307,000	92% (\$1,203,000)
N FINDLAY – FLEWELLEN 138 KV	Rebuild line with higher-capacity conductor	\$8,423,000	100% (\$8,423,000)
FLEWELLEN – BAKER 138 KV	Rebuild line with higher-capacity conductor	\$14,373,000	100% (\$14,373,000)
CEREDO GENERATING STATION – KRS II KOLOA SOLAR 138 KV	Rebuild line with higher-capacity conductor	\$2,229,000	95% (\$2,123,000)
VALEMOUNT 138/1 kV Autotransformer	Replace transformer with higher-rated unit	\$5,348,000	12% (\$634,000)
	Total allocated to project		\$33,401,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.